Savitribai Phule Pune University, Pune

Department of Electronic & Instrumentation Science

A PLE Report on-

W. A. V. E – Wearable Automated Visual Emitter - A Motion-Controlled Traffic Glove (prototype).

By

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(2k24E06)

Under the Guidance of

Asst. Prof. Monica Kamtamkar

In Partial Fulfilment of

MSc – I (Electronic Science)

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My classmates have been of great help to me during the project work.

Date:

Place:

CERTIFICATE

This is to certify that,

Denalven Sawian

Has successfully completed the analysis and the design of M.Sc. – I project and has submitted the Preliminary project report on,

W. A. V. E – Wearable Automated Visual Emitter – A Motion-Controlled Traffic Glove (prototype).

Under the guidance of Department of Electronic and Instrumentation Science, towards the partial fulfillment of Master's degree in Electronic Science under Savitribai Phule Pune University, Pune in the year 2024-25.

Head
Department of Electronic and Instrumentation Science
Savitribai Phule Pune University.

Project Guide

Asst. Prof. Monica Kamtamkar.

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ABSTRACT

Traffic management in urban and rural areas usually depends on human action, especially in locations where automated traffic signals are absent or not functioning. In rural regions, traffic police mostly make use of hand gestures and whistles to direct vehicles, which may not be reliable at all times, particularly during low visibility conditions. The Wearable Automated Visual Emitter (W.A.V.E) is a motion-controlled glove prototype designed to assist traffic personnel by providing a dynamic visual signalling system. Equipped with a microcontroller and powered by a rechargeable battery, the glove responds to specific hand gestures to enhance traffic control efficiency, especially useful in locations lacking traditional infrastructure or during traffic light outages.

I) INTRODUCTION

1.1) Background & motivation:

Being from a semi – developed city like Shillong where there is a lack in proper infrastructure highlights the inefficiency of the traditional traffic management. Since traffic personnel usually wear white gloves while directing traffic an idea struck as to make an electronic glove that could work as a miniaturized traffic light tool that would aid greatly in traditional traffic management. Efficient traffic control is vital to ensure road safety and smooth transportation flow.

1.2) Problem Statement:

In cities like Shillong, and other remote regions, traffic personnel rely heavily on hand signals and whistles, which may not always be efficient. Moreover, urban areas frequently experience disruptions due to power outages, accidents, or emergency situations. To address these challenges, the W.A.V.E glove prototype was developed as a portable, motion-controlled signalling device. Designed for quick adaptability, it enhances visibility by integrating LEDs into a glove, ensuring that traffic personnel can efficiently signal directions even in demanding conditions.

II) LITERATURE REVIEW

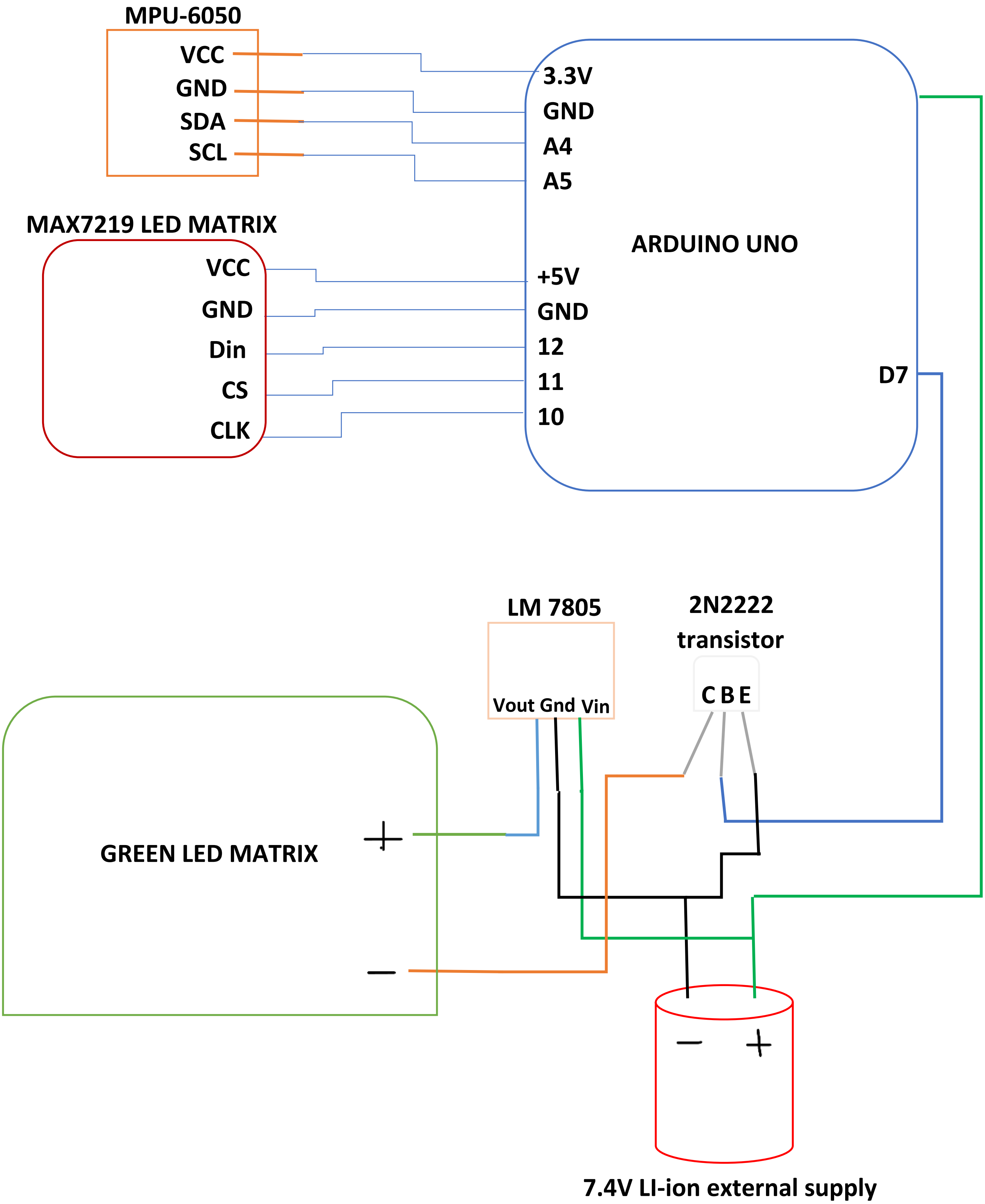
Accurate data on the exact percentage of semi-urban cities in India lacking traffic lights is not readily available. However, it's recognized that many smaller towns and semi-urban areas often operate without such infrastructure, relying instead on manual traffic management. Regarding traffic accidents, intersections without traffic signals, particularly those managed manually or uncontrolled, have been associated with higher casualty rates. In 2022, India reported 2,421 fatalities at police-controlled intersections, exceeding the 2,238 fatalities at signalized intersections. Ref. (I)

Although accurate statistics on the proportion of rural areas lacking traffic lights are scarce, it is known that most rural and semi-urban regions function without such facilities, using manual traffic control or uncontrolled intersections instead. This absence of infrastructure results in increased accident rates in these areas. For e.g. in 2015, 39% of all road fatalities happened in urban regions, suggesting that the other 61% happened in rural regions. Ref. (II)

These figures highlight the urgent need for improved traffic management solutions in India's rural areas to enhance road safety and accommodate the ongoing urbanization trend.

III) SYSTEM DESIGN AND METHODOLOGY

3.1) BLOCK DIAGRAM:



3.2) HARDWARE COMPONENTS:

1. Arduino Uno

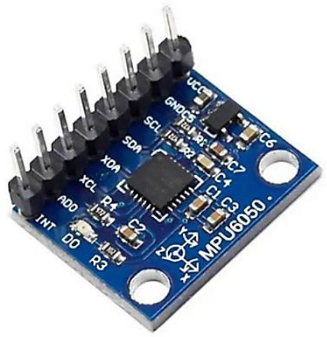
- The Arduino Uno serves as the core microcontroller of the prototype. It processes motion sensor data, interprets the hand gestures, and controls the LED matrices accordingly. The board features an ATmega328P microcontroller, digital and analog I/O pins, and operates at 5V with a clock speed of 16 MHz
- Microcontroller: ATmega328P (8-bit AVR family).
- Operating Voltage: 5V.
- Recommended Input Voltage: 7-12V.
- Input Voltage Limits: 6-20V.
- Analog Input Pins: 6 (A0 – A5).
- Digital I/O Pins: 14 (6 provide PWM output).

- DC Current on I/O Pins: 40 mA.
- DC Current on 3.3V Pin: 50 mA.
- Flash Memory: 32 KB (0.5 KB for Bootloader).
- SRAM: 2 KB.
- EEPROM: 1 KB.
- Frequency (Clock Speed): 16 MHz

2. MPU-6050 (Motion Sensor)

The MPU-6050 is a 6-axis motion tracking sensor that combines a 3-axis accelerometer and a 3-axis gyroscope. It detects tilt, orientation, and movement, allowing the glove to distinguish between different hand gestures. The I2C communication interface is used to transfer sensor data to the Arduino.

- Operating Voltage: 3V.
- Accelerometer range: ±2g, ±4g, ±8g, ±16g.
- Gyroscope range: ±250°/s, ±500°/s, ±1000°/s, ±2000°/s.
- Communication Interface: I2C serial interface with a maximum clock frequency of 400kHz.



- INT: Interrupt digital output pin.
- AD0: I2C Slave Address LSB pin. This is 0th bit in 7-bit slave address of device. If connected to VCC then it is read as logic one and slave address changes.
- XCL: Auxiliary Serial Clock pin. This pin is used to connect other I2C interface enabled sensors SCL pin to MPU-6050.
- XDA: Auxiliary Serial Data pin. This pin is used to connect other I2C interface enabled sensors SDA pin to MPU-6050.
- SCL: Serial Clock pin. Connect this pin to microcontrollers SCL pin.
- SDA: Serial Data pin. Connect this pin to microcontrollers SDA pin.

- GND: Ground pin. Connect this pin to ground connection.
- VCC: Power supply pin. Connect this pin to +5V DC supply.

3. MAX7219 LED Matrix

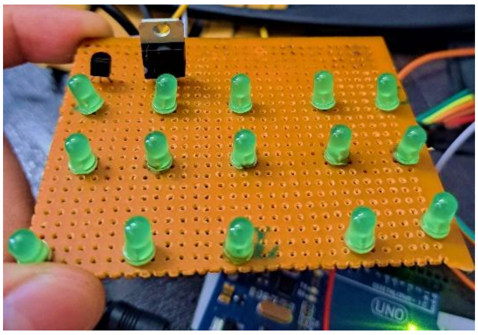
- The MAX7219 is an 8x8 matrix LED display module that is used to display the red STOP signal and work in coordination with the hand gestures. There are 4 of these matrix modules which are mounted closely to each other in the palm side of the glove. The MAX7219 operates on SPI protocol, requiring three control pins (DIN, CS, CLK) to communicate with the Arduino.
- Wide Operating Voltage: 3.7 to 5.3 V
- Input Current: 320 mA.
- 8 x 8 dot matrix Display.
- Serial Interface Control: Allows for control using a simple serial communication protocol, minimizing the number of microcontroller pins needed.

- Cascadable Design: Multiple MAX7219 modules can be daisy-chained to create larger displays without using additional microcontroller pins.
- Individual LED Control: Each of the 64 LEDs in the 8x8 matrix can be individually controlled, enabling the display of a wide range of characters, symbols, and animations.

- Brightness Control: Supports adjustable brightness levels, which can be programmed via software to suit different lighting conditions.
- Low Power Consumption: Efficient power management makes it suitable for portable battery-operated devices.
- Compact Form Factor: Its small size makes it ideal for space-constrained applications.

4. Green LED Matrix with LM7805 and 2N2222 Transistor**.**

An additional green LED matrix of around 15 LEDs is manually soldered to represent the GO signal. The LM7805 voltage regulator ensures a stable 5V power supply to the LEDs, while the 2N2222 transistor acts as a switching component, controlling the matrix based on Arduino signals. The transistor is used to handle the current load efficiently and prevent damage to the microcontroller.

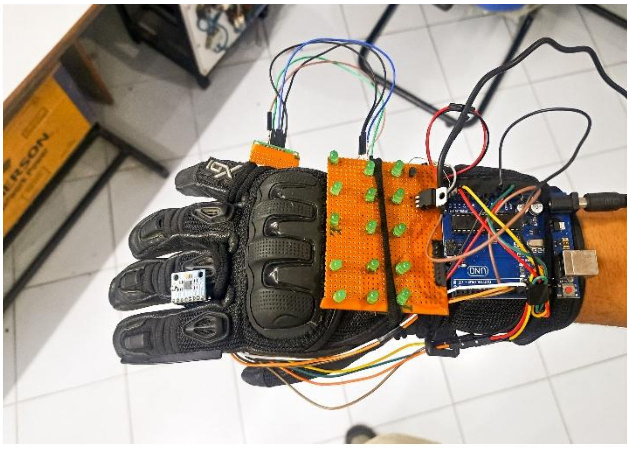


5. 7.4V Li-ion Rechargeable Battery (3000mAh) with Battery Management System (BMS)

The glove is then powered by a 7.4V lithium-ion battery pack, ensuring long-lasting operation. A Battery Management System (BMS) is incorporated to regulate charging, prevent over-discharge, and protect against overheating, enhancing battery safety and longevity.

6. Glove

As this project is a prototype, the glove used in the model was a bike glove and all components were mounted on the glove, it had thick padding and more layers which made the overall weight heavier, since it also had Velcro straps, they were used to place the external power supply and make the model look attractive.



3.3) WORKING PRINCIPLE

The W.A.V.E glove prototype operates by analysing hand gestures and activating the respective LED signals:

- STOP Signal (Red LED Matrix ON) – When the glove is in an upright position with the palm facing outward, the MPU-6050 detects a high Y and Z -axis value. The Arduino processes this data and triggers the MAX7219 LED matrices, illuminating the red stop signal.
- GO Signal (Green LED Matrix ON) – When the hand is flipped downward and showing the front side of the hand, the MPU-6050 registers the change in the Y and Z axis values while also detecting the change in the x axis as the hand approaches to the left. The Arduino interprets this as a 'go' gesture and activates the green LED matrix through the 2N2222 transistor, signalling the traffic to proceed.
- Neutral Position (Both OFF) – When the hand is in a relaxed position, (i.e. not facing up or down) the sensor detects mid-range values across all axes, signalling the Arduino to turn off both LED indicators, ensuring no unwanted signals are displayed.

This design ensures real-time gesture detection and response, enhancing traffic personnel’s ability to control traffic efficiently and clearly.

3.4) FLOW DESIGN

Hand movement up (Stop position)

MPU-6050 senses motion

ARDUINO

Matrix7219 LEDs (RED)-ON

Hand movement down + left (GO position)

MPU-6050 senses motion

ARDUINO

Green LEDs-ON

Hand Relaxed (Neutral position)

Both LEDs-OFF

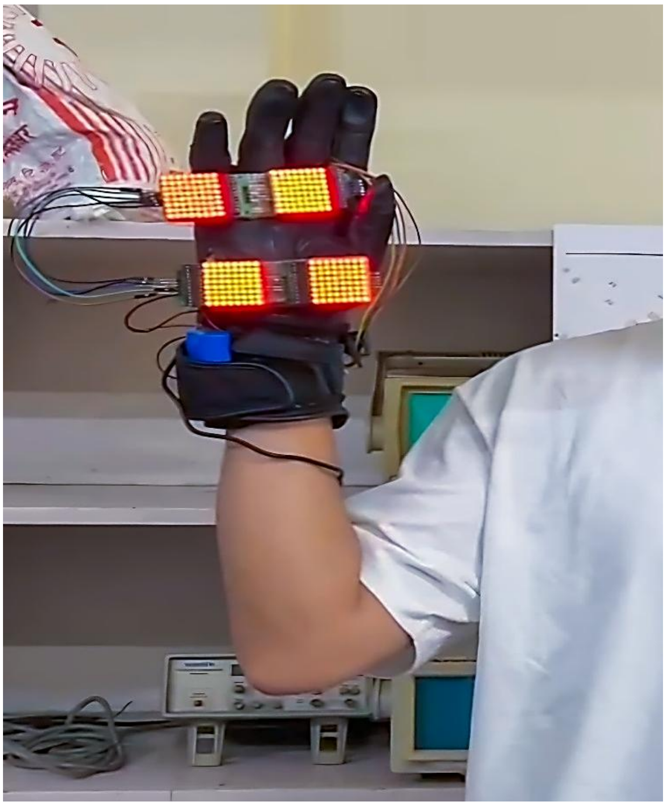
3.5) HARDWARE SETUP

The following table represents the different pin connections made between the MAX7219 LED matrix and the MPU-6050 with the Arduino.

Component Pin	**Arduino Pin**	**Description**
MAX7219 DIN	12	Data input
MAX7219 CS	11	Chip select
MAX7219 CLK	10	Clock signal
MAX7219 VCC	+5V	Power supply
MAX7219 GND	GND	Ground
MPU-6050 VCC	3.3V	Power supply
MPU-6050 GND	GND	Ground
MPU-6050 SDA	A4	I2C Data Line
MPU-6050 SCL	A5	I2C Clock Line

The Green led matrix is soldered on a zero PCB where 15 green LEDs are connected in parallel. The lm7805 takes input from the 7.4V external supply and output is connected to the positive terminal of the LED matrix, the negative terminal is connected with the collector of the 2N2222 transistor with emitter grounded and Digital pin 7 of the Arduino is connected to the base.

IV) RESULT & OUTPUT



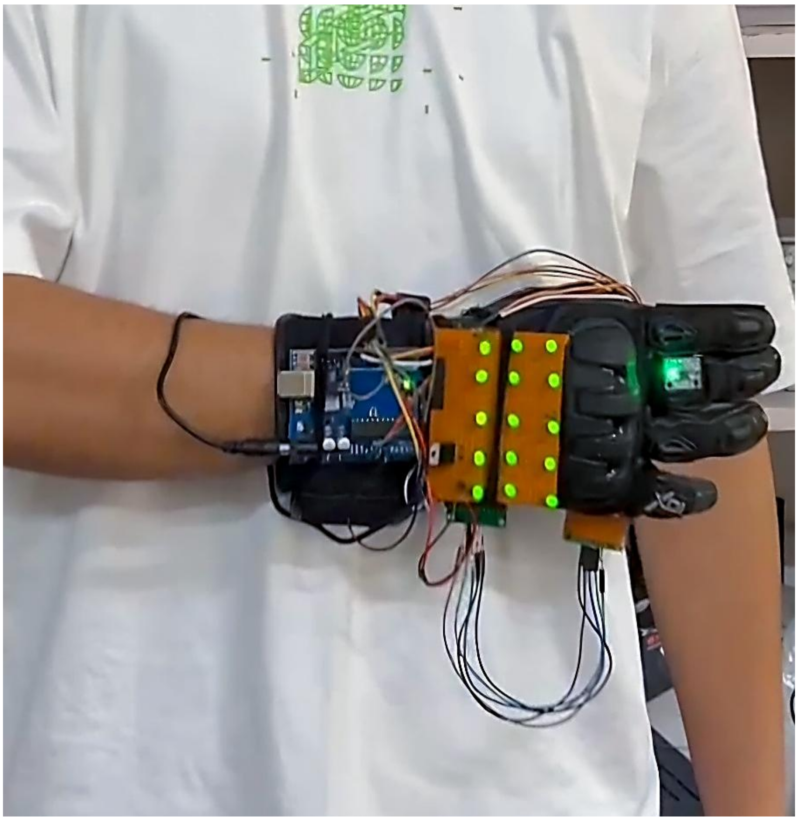
a) Hand at STOP position (raised up) turns on the red LEDs.

b) Glove prototype successfully distinguished the different hand gestures.

c) Hand at GO position (left + downward) turns on the green LEDs.

d) The glove’s motion detection accuracy ensured instant signal switching.

e) Power consumption was minimal, allowing extended use on a single battery charge.



4.1) SOFTWARE:

Arduino IDE

Code:

```
#include <Wire.h>
#include <I2Cdev.h>
#include <MPU6050.h>
#include <LedControl.h>

#define GREEN_MATRIX_PIN 7 // Transistor control for Green Matrix

// MAX7219 Setup
int DIN  = 12;
int CS   = 11;
int CLK  = 10;
int numDevices = 4;  // Number of MAX7219 modules

MPU6050 mpu;
LedControl lc = LedControl(DIN, CLK, CS, numDevices - 1);

int16_t ax, ay, az;
int16_t gx, gy, gz;

struct MyData {
  byte X;
  byte Y;
  byte Z;
};

MyData data;

void setup() {
    Serial.begin(9600);
    Wire.begin();
```

```
  mpu.initialize();
  pinMode(GREEN_MATRIX_PIN, OUTPUT); // Set Green Matrix Pin as OUTPUT

  for (int i = 0; i < numDevices; i++) {
    lc.shutdown(i, false);  // Wake up each MAX7219
    lc.setIntensity(i, 1);  // Set brightness level (0-15)
    lc.clearDisplay(i);     // Clear display
  }
}
void loop() {
  // Read MPU6050 Accelerometer & Gyroscope Data
  mpu.getMotion6(&ax, &ay, &az, &gx, &gy, &gz);

  // Map values for better usability
  data.X = map(ax, -17000, 17000, 0, 255);
  data.Y = map(ay, -17000, 17000, 0, 255);
  data.Z = map(az, -17000, 17000, 0, 255);

  // Print values every 500ms for debugging
  static unsigned long lastPrint = 0;
  if (millis() - lastPrint > 500) {
    Serial.print("X = ");
    Serial.print(data.X);
    Serial.print(" | Y = ");
    Serial.print(data.Y);
    Serial.print(" | Z = ");
    Serial.println(data.Z);
    lastPrint = millis();
  }

  // **GESTURE 1: STOP (RED MATRIX ON)**
  if (data.Y > 145 && data.Z > 140) {
    Serial.println("STOP GESTURE - Turning ON RED MATRIX");
    for (int dev = 0; dev < numDevices; dev++) {
```

```
            for (int row = 0; row < 8; row++) {
                lc.setRow(dev, row, 0xFF);  // Turn on Red Matrices
            }
        }
        digitalWrite(GREEN_MATRIX_PIN, LOW); // Ensure Green Matrix is OFF
    }
    // **GESTURE 2: CALLING / GO (GREEN MATRIX ON)**
    else if (data.X < 50 && data.Y < 135 && data.Z > 120) {
        Serial.println("GO GESTURE - Turning ON GREEN MATRIX");
        digitalWrite(GREEN_MATRIX_PIN, HIGH);
        for (int i = 0; i < numDevices; i++) {
            lc.clearDisplay(i);  // Ensure Red Matrices are OFF
        }
    }
    // **GESTURE 3: RESET TO NEUTRAL (TURN OFF BOTH)**
    else if (data.Y < 80) {
        Serial.println("Neutral Position - Turning OFF BOTH");
        digitalWrite(GREEN_MATRIX_PIN, LOW);
        for (int i = 0; i < numDevices; i++) {
            lc.clearDisplay(i);
        }
    }
}
```

The software for the W.A.V.E glove is designed to interpret motion data from the MPU-6050 sensor and control the LED matrices in real-time based on hand gestures. The program follows these key operations:

- The MPU-6050 continuously measures acceleration and gyroscopic values, sending data to the Arduino via I2C.
- The Arduino processes sensor readings and maps the Y and Z-axis values to determine the glove's orientation.
- Based on the analyzed data, specific gestures trigger different responses:
    - STOP Gesture: When the palm is upright, the Y-axis value crosses a threshold, activating the red LED matrices via the MAX7219 driver.
    - GO Gesture: When the hand is tilted forward, the sensor detects lower Y-values, prompting the Arduino to switch on the green LED matrix by controlling the 2N2222 transistor.
    - Neutral Position: If no defined gesture is detected, the Arduino turns off both LED indicators, preventing unintended signals.

4.2) LIMITATIONS & CHALLENGES

- Bulkiness – The combination of multiple components makes the glove somewhat bulky, affecting ease of use.
- Microcontroller Performance – The Arduino Uno has limited processing power; a more advanced microcontroller could improve response times and add wireless capabilities.
- Power Efficiency – The battery life is dependent on LED brightness and frequency of use; optimizing power consumption could enhance operational duration.
- Sensor Accuracy – The MPU-6050, while effective, may experience drift or calibration issues, affecting long-term reliability.
- Durability – Exposure to harsh weather conditions could impact the glove’s performance, necessitating a more robust enclosure.

V) CONCLUSION & FUTURE SCOPE

Future scope for the W.A.V.E glove prototype could include:

- Lightweight and Compact Design – Integrating smaller, embedded components would reduce the number of wires removing the bulkiness and improve wearability and weight.
- Advanced Microcontroller – Upgrading to a more powerful microcontroller could enhance processing speed and enable wireless communication for remote monitoring.
- Enhanced Power Management – Implementing low-power components and an optimized power management system could extend battery life.
- Better Sensor Integration – Using a more precise and stable motion sensor could improve gesture recognition accuracy.
- Wireless Connectivity – Adding Bluetooth or RF communication would allow the glove to interact with smart traffic systems and remote-control units.
- Weatherproofing – A more durable, weather-resistant design would ensure reliable operation in harsh conditions.
- Other Applications – The technology could be adapted for emergency signalling, assisting visually impaired pedestrians, or integrated into smart city traffic management systems.

As the world moves toward smart and adaptive traffic solutions, what was merely an idea turned into a physical working prototype, the **Wearable Automated Visual Emitter (W.A.V.E) glove** stands as a symbol of innovation in manual traffic control. It is a combination of gesture recognition and visual signaling that provides a cost-effective, portable, and efficient means of managing traffic where traditional systems fail. Although still in its prototype phase, its potential to increase road safety and transform traffic signaling is undeniable. With further advancements, it could evolve into a seamless, integrated tool—not just for traffic control, but for emergency responders, industrial workers, and more. The future is bright, and the best is yet to come.

5.1) REFERENCES

I) The Times Of India - https://timesofindia.indiatimes.com/india/more-crashes-deaths-at-cop-manned-junctions-than-at-traffic-signal-spots/articleshow/104898147.cms#:~:text=What's%20more%2C%20fatalities%20at%20police,traffic%20light%20signals%20(2%2C238).

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